Information Engineering and Technology Faculty
German University in Cairo

Enhancing LoRa LPWAN Performance: Machine Learning and AI Approaches

Bachelor Thesis

Author: Belal Hesham Mohamed Mohamed
Supervisors: Dr.Minaar El Aasser
Submission Date: 19 May, 2024

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This is to certify that:

(i) the thesis comprises only my original work toward the Bachelor Degree

(ii) due acknowlegement has been made in the text to all other material used

Belal Hesham Mohamed Mohamed
19 May, 2024

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First and foremost, I thank God for blessing my life and enabling me to succeed in my studies. I am grateful to my supervisor, Dr.Minaar El Aasser, for her direction, supervision, and readiness to provide suggestions. I'd want to thank all my friends who assisted me with the project and provided valuable notes and suggestions. Finally, I want to express my gratitude to my family for their unwavering support in getting me to where I am now.

Abstract

As technology progressed, alternatives to Wi-Fi and cellular networks, which overcome their drawbacks were identified. They absorb too much power and progressively deplete battery life. Wi-Fi has limited coverage and inadequate security, making it a significant disadvantage. LoRa was designed to carry data across a large coverage range with minimal power. A dataset of LoRa parameters is simulated on Omnett++ using ADR to assign SF values for each user. Results are then compared with various Machine Learning approaches. This is due to ADR's limitations, including its inability to handle dynamic objects. The Machine Learning algorithms employed include KMeans Clustering, KNN, Naive Bayes, Random Forest, Support Vector Machine, and GMM. Finally, the assessment metrics for all procedures are calculated, and the outcomes are analyzed where the aforementioned supervised methods, namely, KNN, Naive Bayes, Random Forest, and Support Vector Machine show accuracy above 99%, with high precision, recall, and F1 scores. K-Means Clustering and Gaussian Mixture Models (GMM) are effective tools for clustering tasks, as silhouette scores fall in the range of approximately 0.67 to 0.68 and Davies Bouldin indices indicating well-separated clusters.

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Chapter 1

Introduction

1.1 Motivation

For the wireless world to function perfectly, people now need to have wireless communication. This is where wireless communication protocols come in handy. They allow for connectivity between different devices including smartphones, laptops, and smart homes within a building or area. Apart from just being able to be used in our homes, these protocols ensure that even industries can benefit from them through things like automation systems which require sensors remotely placed around a plant for monitoring purposes among others. LoRa technology which stands for long range is another popular option when setting up networks over long distances because it consumes less power than both WiFi and cellular networks do, while at the same time being able to cover very vast areas. Many IoT projects use this type of protocol for connecting various gadgets together thereby forming what can be termed as intelligent networks.

1.2 Aim of the Thesis

The primary aim of this thesis is to address the issue of selecting the best spreading factor (SF) in LoRa networks. Spreading factor has a significant impact on the data rate and transmission range. This thesis intends to enhance the performance of LoRa networks by using machine learning techniques for spreading factor optimization. It details the research objectives and specific aims by underlining that machine learning algorithms will be applied to increase network efficiency and scalability.

1.3 Thesis Outline

This section provides an outline of the thesis' structure and what each chapter covers. It briefly discusses a variety of topics, including background information about LoRa technologies, their features, architecture, and even how machine learning techniques can be

used in spreading factor assignments. It also mentions some other things, such as relevant studies that were conducted prior to this one, algorithms that were chosen along with their respective datasets, results obtained from performance evaluations, and concluding remarks about future research possibilities.

Chapter 2

Background

Here, the building blocks and technologies related to LoRa are reviewed. The LoRa device technology, designed for giant-scale IoT deployment, gives a unique combination of super low power consumption along with long-range capabilities. Key elements of LoRa communication, such as bandwidth, spreading factor, and code rate which considerably influence performance and suitability for various IoT applications are explained. Besides, LoRa with other wireless technologies is compared, in particular, discussing its advantages regarding range and power consumption. Then the architecture of the LoRa system is explained, including the role of end devices, gateways, network servers, and application servers for enabling smooth communication and data management in LoRa networks. Lastly, the contribution of artificial intelligence, and machine learning in optimizing LoRa network performance is mentioned, including network optimization, traffic analysis, security, detection of anomalies, and energy consumption optimization.

2.1 What Is LoRa?

LLoRa, short for Long Range, is a wireless communication technology ideal for massive-scale IoT deployments, combining ultra-low power consumption with impressive long-range capability. The LoRa stack protocol consists of a physical layer, which has been enhanced with a MAC layer, resulting in LoRaWAN (Long Range Wide Area Network). LoRaWAN enables communication between sensors and the cloud in IoT environments, making it especially beneficial for applications in smart cities, agriculture, and various other sectors.

2.1.1 LoRa Parameters

- Bandwidth: In LoRa (Long Range) communication, bandwidth refers to the range of frequencies allocated for transmitting signals. It is a configurable parameter that influences the width of the frequency channel used by LoRa devices. Common bandwidth options include 125 kHz, 250 kHz, and 500 kHz.

- Spreading Factor: In LoRa (Long Range) communication, spreading factor refers to a key parameter that influences the modulation scheme and signal spreading of transmitted data. LoRa devices use spread spectrum modulation, and the spreading factor determines how wide the transmitted signal is spread across the frequency spectrum. Spreading factors are represented by numerical values (e.g., SF7, SF8, SF12), with lower values indicating wider spreading and higher values indicating narrower spreading.
- Code Rate: In LoRa (Long Range) communication, code rate refers to the error-correcting capability applied to the transmitted data. It is represented as a ratio, such as 4/5 or 4:5, where the first number indicates the amount of error-correcting code added, and the second number represents the total code bits transmitted.

2.2 LoRa vs Other Wireless Technologies

Bluetooth is good for short-range, low-power situations, encouraging perfect communication in the scenario of personal area networks, but it may be susceptible to interference and limited in the number of connected devices. Wi-Fi provides high data rates at a moderate range, making it ideal for high-bandwidth applications but consumes more power and may be congested in environments with a high number of users. Cellular networks provide broad coverage with high data rates but can be power-intensive for connected devices and may have subscription costs and lack of coverage in remote areas. On the other hand, loRa marks itself because of its long-range capabilities and low power consumption, making it a necessity for IoT applications that require connection over long distances. As can be seen from Figure 2.1, loRa consumes less bandwidth than Wi-Fi and cellular and can cover longer range.

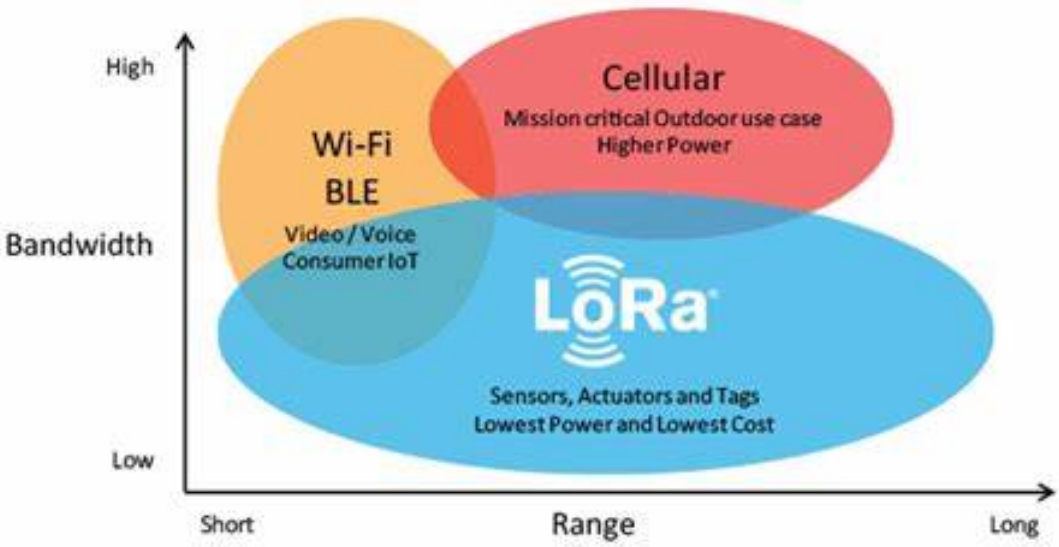

Figure 2.1: LoRa vs Other Wireless Technologies

2.3 LoRa architecture

The LoRa architecture comprises several key components that collectively enable seamless communication within LoRa networks. These components include end devices, gateways, network servers, and application servers, as shown in Figure 2.2 each playing a critical role in the network ecosystem.

- End Devices: They are typically sensor nodes or devices that collect data where these sensors can measure various parameters such as temperature, humidity, etc.

- Gateways: Act as a bridge between end devices and the network server where they collect data from end devices and forward it to the network server.

- Network Server: Plays an important role in managing and coordinating communication between end devices and application servers.

- Application Server: Plays a crucial role in managing and processing data transmitted by LoRa-enabled devices. It serves as the endpoint where data from LoRa devices is received, processed, and often integrated into higher-level applications or systems.

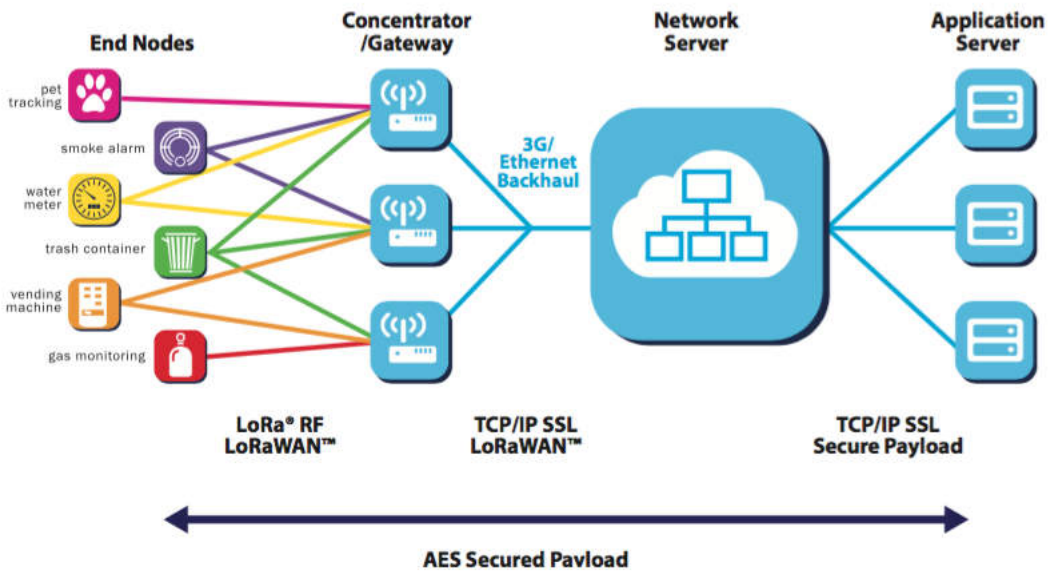

Figure 2.2: LoRa Architecture

2.4 LoRa Stack Protocol

LoRa stack protocol comprises two layers: the LoRa physical layer and the LoRaWAN MAC layer, as shown in Figure 2.3 . Together, these layers form the backbone of the LoRa communication protocol, it enables efficient and long-range communication for Internet of Things (IoT) applications.

2.4.1 LoRa (Physical Layer)

In the context of wireless communication, the physical layer defines transmission characteristics such as modulation schemes, which for LoRa is usually Chirp Spread Spectrum (CSS). This layer, essentially, controls how data is transmitted over the air-waves.

2.4.2 LoRaWAN (MAC Layer)

LoRaWAN (MAC Layer) is a communication protocol designed for low power wide-area networks, made specifically for IoT applications. The signal from the low-power end device to the gateway may be extended further in the communication through the LPWAN. LoRaWAN regulates three classes of devices: Class A, Class B, and Class C. Class A devices are more frequent than the other; they operate based on a "listen-before-talk" protocol. After sending data, they open their receive windows. Class B devices have scheduled receive slots, which allows the real-time supply of more deterministic communication at a higher cost in terms of power. Class C devices keep their receive windows open most of the time, which allows very low latency.

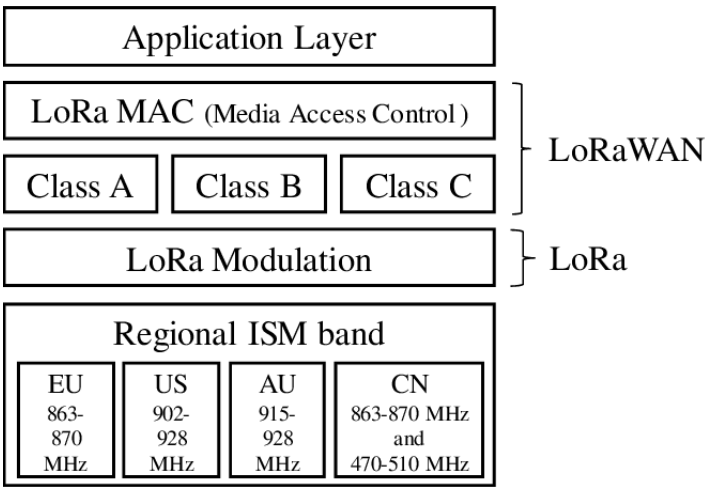

Figure 2.3: LoRa Stack Protocol

2.5 LoRa and Spreading Factor

LoRa SF is the most vital element of the LoRa network because it influences both the coverage range and the data rate of communication. LoRa devices use the modulation of the spreading of signals to achieve different degrees of coverage and data transfer efficiency. SF selection directly affects the trade-off between communication range and data rate.

2.5.1 High SF vs Low SF

The selection of spreading factors in LoRa is a trade-off of some factors, including the data rate, communication range, and power consumption. Low spreading factors offer high data transfer rates, which are suitable for applications requiring real-time data or high bandwidth communication. However, high speed comes at the cost of a less extended communication range and more power consumption, which reduces their utility in applications that require long-range connectivity or energy efficiency. On the other hand, high spreading factors offer long communication ranges at the cost of consuming less power as shown in Table 2.1, which makes them viable for applications in which the geographic area is broad or energy efficiency is a constraint.

Spreading Factor (For UL at 125 KHz)	**Bit Rate**	**Range (Depends on Terrain)**	**Time on Air (for an 11-byte payload)**
SF10	980 bps	8 km	371 ms
SF9	1760 bps	6 km	185 ms
SF8	3125 bps	4 km	103 ms
SF7	5470 bps	2 km	61 ms

Table 2.1: Spreading Factor, Bit Rate, Range, and Time on Air for an 11-byte payload

2.6 Adaptive Data Rate(ADR)

ADR refers to a dynamic and automated mechanism designed to optimize the data rate and transmission parameters of LoRa devices within a LoRaWAN (LoRa Wide Area Network) network.ADR is specifically employed to enhance the overall efficiency and performance of communication between IoT devices and the network server.The primary goal of ADR is to adapt the data rate of LoRa devices based on the quality of the radio link between the device and the gateway. As environmental conditions and distances between devices and gateways can vary, ADR helps maintain an optimal balance between

power consumption, communication range, and data transfer speed.The LoRa network server analyzes the received signal strength, signal-to-noise ratio, and other metrics to determine the most suitable data rate for a particular device. By dynamically adjusting the data rate, ADR aims to maximize the communication range when conditions permit and minimize power consumption when devices are in proximity to gateways.

2.6.1 Determining the Data Rate

The end device's application sends a message up through the gateway, which simply passes the message along without acting on the data. The gateway in a LoRaWAN network is a simple, low-cost device that converts LoRaWAN packets into IP packets which can be sent over a secure backhaul to the network server. These IP packets include a small amount of metadata about the reception time and signal strength. Based upon the strength of the received signal, the network server determines what the optimal node data rate should be (that is, the spreading factor).The Media Access Controller, also called the MAC layer, of the network server, talks to the same layer in the LoRaWAN stack of the end-node. A MAC command is then issued from the server based upon its global view of the strength of the signals received from all gateways. The data rate that the node should use, is sent back to the device from the server through the gateway with the best signal strength.

2.7 Artificial Intelligence and Machine Learning

2.7.1 Artificial Intelligence(AI)

Artificial Intelligence refers to the development of computer systems capable of performing tasks that typically require human intelligence. These tasks include learning, reasoning, problem solving, perception, natural language understanding, and decision making.

Types of AI

- Narrow or Weak AI: Weak AI is designed and trained for a specific task or a narrow set of tasks. It operates within a limited context and does not possess general intelligence.

- Strong AI: Strong AI refers to artificial intelligence that has the ability to understand, learn, and apply knowledge across a wide range of tasks, similar to human intelligence. Achieving strong AI is a theoretical goal and has not been fully realized yet.

Drawbacks of AI

Artificial Intelligence presents much promise for a high-impact revolution in so many aspects of human life, but it is associated with a number of significant drawbacks as well. One major concern is the loss of jobs, as AI and automation technologies increasingly replace human labor, leading to unemployment and economic inequality. Further ethical issues are linked to the use of AI in decision-making processes, particularly in serious situations such as healthcare and the justice system: errors or biases can have tremendous negative consequences in the lives of people.

2.7.2 Machine Learning

Machine Learning is the field of AI that aims at creating algorithms and statistical models for the use of computers to learn from data. Instead of being explicitly programmed for a task, machine learning systems are worked out on data and improve over time.

Types of Machine Learning

- Supervised Machine Learning:In supervised machine learning, the algorithm is learned from a labeled dataset. The data in the dataset consists of input features as well as corresponding output labels. The task aims to learn the underlying patterns or relationships between input features and output labels in order to enable accurate prediction or classification on new, unseen data. It ranges from regression tasks, where the model predicts a continuous outcome, to classification tasks, which place the data points into previously defined categories or classes. Common supervised algorithms are linear regression, logistic regression, decision trees, random forests, support vector machines, and neural networks. Success in supervised learning highly depends on the quality and representativeness of labeled data, as well as the choice of appropriate features and algorithms.

- Unsupervised Machine Learning:In unsupervised machine learning, algorithms are trained on unlabeled datasets. No predefined output labels are given. The primary objective is to learn patterns, structures, or relations within the data by the algorithms autonomously. This usually happens in the case where the goal of the work is discovering the underlying structure or distribution of the data. The most common techniques used in unsupervised learning are clustering—in which the data points are grouped with each other based on similarity or proximity—and dimensionality reduction. Popular unsupervised algorithms include k-means clustering, and hierarchical clustering. Unsupervised learning models can commonly be assessed by considering their coherence of the discovered patterns and their downstream use in tasks such as data visualization, anomaly detection, or feature engineering.

- Reinforcement Learning:Reinforcement learning is a type of machine learning where an agent learns to perform actions in an environment. It receives feedback in the

form of rewards or penalties over time as it interacts with the environment and the task at hand is to learn a policy that maximizes cumulative reward.

Applications Of Machine Learning

Machine learning is regarded as a revolutionary and efficient tool with applications across various fields and domains. In healthcare, machine learning assists in diagnosis, personalized treatment, and finding new drugs, and all these can be a path toward significantly better patient outcomes and reduction of healthcare costs. In finance, machine learning algorithms are used in analysis of market trends, detection of fraud, and optimization of trading and much better in risk management; investment decisions will also be supported. In marketing, it plays a role for targeted advertising, customer segmentation, and recommendation systems, enhancing customer engagement and loyalty. Machine learning in the case of autonomous vehicles is used to achieve real-time navigation, object detecting, decision-making, and it is the step toward self-driving cars and improving road safety. Machine learning in natural language processing will enable language translation, sentiment analysis, and speech recognition, which are driving innovations into virtual assistants, chatbots, and language understanding systems.

2.8 Machine Learning and loRa

- Network Optimization: Implementing machine learning algorithms to dynamically adjust parameters such as spreading factor, data rate, and transmission power based on network conditions. This enhances efficiency and extends the network's coverage.
- Traffic Analysis: Employing machine learning models to analyze data patterns, identify anomalies, and optimize communication strategies.
- Security and Anomaly Detection:Implementing anomaly detection algorithms to identify unusual patterns in communication that may indicate security threats.
- Energy Consumption Optimization: Developing machine learning models to predict energy usage patterns and optimize the operation of battery-powered devices. This can extend the lifespan of devices and improve overall energy efficiency.

2.9 Machine learning techniques

The six machine learning algorithms that have been put into practice will be covered in this section, along with a step-by-step description, each algorithm's benefits, and drawbacks.

2.9.1 KNN

How KNN Works

- Classification:
- Step 1: Identify k-nearest neighbors of new data point.
- Step 2: Count occurrence of each class label among neighbors.
- Step 3: Assign most frequent class label to new data point.
- Regression:
- Step 1: Locate k-nearest neighbors of new data point.
- Step 2: Retrieve target values of these neighbors.
- Step 3: Compute average of target values.
- Step 4: Assign average as predicted output for new data point.

Advantages of KNN

- Simple and Easy to Understand
- Non-parametric
- Can Handle Large Datasets
- Accurate and Effective

Disadvantages of KNN

- Sensitive to Outliers
- Computationally Expensive
- Requires Good Choice of K

2.9.2 Naive Bayes

How Naive Bayes Works

- Step 1: Calculate the prior probability of each class based on the frequency of occurrence in the training dataset.
- Step 2: For each feature in the new data point, calculate the likelihood of observing that feature given each class.
- Step 3: Multiply the prior probability of each class by the likelihoods calculated in step 2 to obtain the joint probability for each class.
- Step 4: Normalize the joint probabilities to obtain the posterior probabilities for each class.
- Step 5: Assign the class with the highest posterior probability as the predicted class label for the new data point

  .

Advantages of Naive Bayes

- Simple to Implement
- Very fast
- If the conditional Independence assumption holds, it could give great results.

Disadvantages of Naive Bayes

- Conditional Independence Assumption does not always hold. In most situations, the feature show some form of dependency.
- Zero probability problem

2.9.3 K-Means Clustering

How K-Means Works

- Step 1: Initialize cluster centroids randomly or using a predefined method.
- Step 2: Assign each data point to the nearest cluster centroid based on a distance metric (e.g., Euclidean distance).

- Step 3: Update the cluster centroids by calculating the mean of all data points assigned to each cluster.
- Step 4: Repeat steps 2 and 3 until convergence criteria are met (e.g., no change in cluster assignments or centroids).
- Step 5: The final cluster centroids represent the cluster centers, and each data point is assigned to the cluster with the nearest centroid.

Advantages of K-Means

- Relatively simple to implement.
- Scales to large data sets.
- Guarantees convergence.
- Easily adapts to new examples.

Disadvantages of K-Means

- Being dependent on initial values.
- Clustering data of varying sizes and density.

2.9.4 Random Forest

How Random Forest Works

- Step 1: In the Random forest model, a subset of data points and a subset of features is selected for constructing each decision tree. Simply put, n random records and m features are taken from the data set having k number of records.
- Step 2: Individual decision trees are constructed for each sample.
- Step 3: Each decision tree will generate an output.
- Step 4: Final output is considered based on Majority Voting or Averaging for Classification and regression, respectively.

Advantages of Random Forest

- It can be used in classification and regression problems.
- It solves the problem of overfitting as output is based on majority voting or averaging.
- It performs well even if the data contains null or missing values.

Disadvantages of Random Forest

- Random forest is highly complex compared to decision trees, where decisions can be made by following the path of the tree.
- Training time is more than other models due to its complexity.

2.9.5 Support Vector Machine

How Support Vector Machine Works

The purpose of SVM is to identify the hyperplane that optimally separates data points from distinct classes in the feature space. In binary classification (two classes), this hyperplane is a line. For multiclass classification, a multidimensional plane is used as in Figure 2.4. SVM not only strives to discover a separating hyperplane, but also to maximize the margin, which is the distance between the hyperplane and the nearest data points in each class. This margin ensures that the classifier is robust and generalizable.SVM formulates an optimization problem to find the hyperplane with the maximum margin. It minimizes the classification error while maximizing the margin.Support vectors are the data points nearest to the separating hyperplane. They are critical for defining the hyperplane and establishing the margin. Only these support vectors influence the decision border, making SVM memory efficient.SVM also includes a regularization parameter (C) that controls the trade-off between maximizing margin and decreasing classification error. A greater value of C allows for a smaller margin and fewer classification errors, whereas a lower value of C results in a bigger margin and maybe more misclassification.Once the best hyperplane has been found, SVM uses it to classify previously undiscovered data points. It allocates them to a class based on which side of the hyperplane they are on.

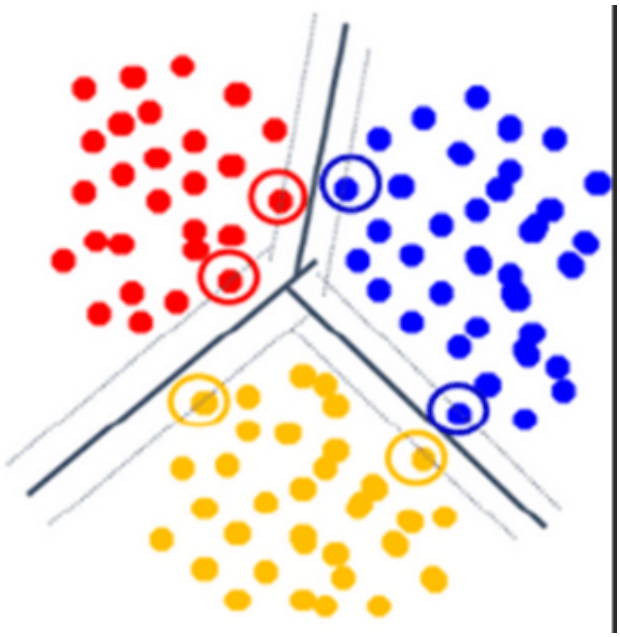

Figure 2.4: Multidimensional classifier

Advantages of Support Vector Machine

- SVM classifiers perform well in high-dimensional space and have excellent accuracy. SVM classifiers require less memory because they only use a portion of the training data.
- SVM performs reasonably well when there is a large gap between classes.
- SVM uses memory effectively.

Disadvantages of SVM

- SVM requires a long training period; as a result, it is not practical for large datasets.
- The inability of SVM classifiers to handle overlapping classes is another drawback.
- The SVM will perform poorly when the number of features for each data point is greater than the number of training data samples.

2.9.6 Gaussian Mixture Model

How Gaussian Mixture Model Works

Consider a bell-shaped curve that shows how the data points are distributed in a single dimension. The curve's variance and mean, or center and spread, respectively, are its

two primary properties. The likelihood of finding a data point at any given value is represented by this curve. We now suppose that the data is composed of multiple bell-shaped curves, each of which represents a distinct group or cluster of data points, rather than just one bell-shaped curve. Every cluster has a unique core and spread that resemble bell curves. In order to best fit the data, we are attempting to determine the location and distribution of these bell-shaped curves. This is made possible via the EM algorithm. This is how it functions:

- **Expectation Step (E-step):** We start with some initial guesses about the positions and spreads of the bell-shaped curves. Then, for each data point, we calculate the probability that it belongs to each cluster based on the current guesses.

- **Maximization Step (M-step):** Using the probabilities calculated in the E-step, we update guesses about the positions and spreads of the bell-shaped curves to better fit the data.

The final model is achieved after we have iterated the E-step and M-step sufficiently times, and estimates of the bell-shaped curves no longer vary significantly. Now, based on their probabilities, we can use the model to determine which cluster each new data point most likely belongs to whenever we receive new data points.

Advantages of GMM

- **Flexiblity:** When it comes to simulating intricate data distributions, GMMs are incredibly versatile. They are appropriate for a number of applications because they can capture a broad range of shapes and structures in the data.

- **GMMs offer soft clustering:** which is different from hard clustering in that each cluster is given a probability of membership rather than a strict assignment. When data points may belong to more than one cluster or when clusters overlap, this can be advantageous.

Disadvantages of GMM

- **Number of Components:**Choosing the right number of components (clusters) for a GMM can be difficult and may call for heuristic methods or domain expertise. Overfitting or underfitting of the data might result from choosing an improper number of components.

- **Starting over Sensitivity:**The initial parameter values that were used to fit the model have an impact on the GMMs. Inadequate starting points may cause convergence to less-than-ideal solutions or perhaps stop there.

2.10 Platform

2.10.1 Python

Python is a widely-used programming language, particularly popular for data science and machine learning applications. One of the key libraries that make Python powerful in these domains is scikit-learn (sklearn). Scikit-learn provides simple and efficient tools for data analysis and modeling, encompassing a wide range of machine learning algorithms for classification, regression, and clustering.

2.10.2 Matlab

MATLAB, short for Matrix Laboratory, is a high-level programming language and environment specifically designed for numerical computing and data visualization. One of its core strengths lies in its advanced graphing capabilities, which allow users to create a wide variety of plots and charts with ease. MATLAB's plotting functions enable the visualization of data in both 2D and 3D, providing tools for line plots, scatter plots, histograms, surface plots, and more. The graphical interface is highly customizable, allowing users to adjust colors, markers, axes, and annotations to produce publication-quality figures.

Chapter 3

Related Work

In this section the relevant work will be discussed and the effect of machine learning on the optimal selection of parameters which will tell how the loRa network will improve and compare these algorithms with ADR and others. Most of algorithms are supervised as supervised learning relies on labeled datasets where the input data is paired with corresponding output labels. Labeled datasets are crucial when it comes to training and evaluating models in a supervised environment. Global application of this approach proves its significance over reinforcement learning strategies which require collecting massive numbers of unlabelled samples or creating appropriate atmospheres. Labeled data is also valuable because machine learning algorithms can predict or classify with increased accuracy by imitating previous examples. Superseding its counterparts like unsupervised learning, which only group similar items together nor reinforcement learning that uses rewards to drive decision-making, supervised learning prevails across different industries due to availability of labeled datasets as well as the ability to repurpose them for developing predictive models.

In [1] a supervised learning method (Random Forest)which outputs a suggested parameter combination for implementation in short time to achieve rapid deployment. It is an attempt to predict performance indicators with a regression model to provide multiple parameters combination. Then all of parameters are sorted based on random forest prediction result. The parameter decision algorithm will return predicted sequence of parameters.

In [2] a supervised machine learning algorithm is implemented where an artificial neural network was built and trained to predict received power then particle swarm algorithm was used to find optimal values of parameters corresponding to max received power. The work aims to maximize link performance by optimizing received power. A data set is generated for each spreading factor to be utilized in building ANN(Artificial Neural Network) to predict loRa network performance. PSO(Particle Swarm Optimization) is utilized to optimize loRa parameters on which received power depends. This cause SNR(Signal to Noise Ratio) and spectral efficiency improvement.

In [3]machine learning algorithm is used where Fuzzy C-means is applied to acquire knowledge about each transmissions setting. This knowledge is used in Q-learning and

markov decision process (MDP) as a state transition matrix where NS(Network Server) has a role to predict new parameters to enhance previous up link data rate. This algorithm improves quality of service (Qos) including data rate, packet delivery rate (PDR), and transmission energy.

In [4] algorithm for problem of SF allocation on basis of K-means clustering is deployed.The algorithm assigns a maximum range of individual SF regions which allows for distinct user distribution where the main objective is to find minimum distance between each point and nearest centroid and assign it to its cluster. Five iteration of K-means are used.

In [5] they aim to implement fuzzy interference system algorithm as controller in IIOT application. The FIS(Fuzzy Interference System) predicts network efficiency tuned to obtain optimized loRa physical parameter under given environment settings where a correlation model is adopted and isolate significant parameters and relation between them. The result of correlation acts as a feedback for designing rules. The rules is applied to FIS(Fuzzy Interference System) and network efficiency is predicted.

In [6] a supervised Ml is used for prediction of ED. It used RSSI fingerprints to locate EDs. Fingerprinting consists of two phase online and offline. In offline measurements are taken while in online location of EDs is predicted using ML. The location is predicted using SVR which find distance error (error between truth location and predicted one). This method locates ED with low average distance error.

In [7]a supervised ML called KNN used for resource allocation where similar nodes are near to each other. After learning process unseen data is fed into KNN algorithm and distance is calculated then decision to which class it should be assigned is made according to pre-dominant class of K nearest points. This algorithm indicates that both energy efficiency and reliability are improved.

In [8] a deep learning method called long short-term memory extend of kalman filter(LSTM-EKF)for collision prediction in LPWAN. The model takes previous information of traffic in loRa system and makes prediction for number of collisions in the future. Collisions occur when there is overlap in reception due to SF collisions, CF collisions, and power collisions. LSTM(Long Short Term Memory) is used to predict future states or observations based on past-data. The output is fed in EKF(Extend of Kalman Filter) framework which help for any errors or uncertainties introduced in prediction process. This can help network administrators to deal with allocation management problems.

In [15] FSVM is an extension of the Support Vector Machine (SVM) algorithm that introduces fuzzy membership functions to handle classification problems with noise or outliers more effectively. It aims to construct a classification hyperplane with the maximum margin between classes while considering the fuzzy membership values of samples, which represent the degree to which a sample belongs to a certain class. FSVM assigns different weights to samples based on their importance and employs machine learning techniques for training and learning.

We have summarized some machine learning algorithms discussed in this section in tables 3.1 , 3.2, 3.3, 3.4, 3.5, 3.6.These tables provide additional details on each algorithm,

including their inputs and outputs, which are essential for understanding their application in loRa.

Topic	Description	Algorithm	Category	Inputs	Outputs	Results
[4]	Algorithm for problem of SF allocation on basis of K-means clustering. The algorithm assigns a maximum range of individual SF regions which allows for distinct user distribution where the main objective is to find minimum distance between each point and nearest centroid. Five iterations of K-means are used.	K-means clustering	Un-supervised ML	Information about locations of nodes and parameters as spreading factor and transmission power.	Cluster assignments for nodes based on similarities in geographical proximity and for each cluster there is a centroid and allocation of spreading factor is done for each cluster and this is done iteratively till convergence.	Better success probability and fair resource distribu-tion.
[3]	ML algorithm where Fuzzy C-means is applied to acquire knowledge about each transmissions setting.This knowledge is used in Q-learning and MDP as a state transition matrix where NS has a role to predict new parameters to enhance previous uplink data rate.	Markov decision process and fuzzy C-means clustering	Supervised ML	Bandwidth, coding rate,spreading factor,metrics to evaluate quality of loRa transmission set-tings,factors related to environment as path loss expo-nent.Markov decision process parameters as transition probabilities and rewards.	Recommended adjusting for loRa transmission based on MDP.Improved data rate and quality of link based on the recom-mendation.	Improving Qos including data rate,PDR and trans-mission energy.

[5]	This paper aims to implement fuzzy interference system algorithm as controller in IIOT application.The FIS predicts network efficiency tuned to obtain optimized loRa physical parameter under given environmental settings where a correlation model is adopted and isolate significant parameters and relation between them.The result of correlation acts as a feedback for designing rules.The rules is applied to FIS and NE is predicted.	Fuzzy interference system.	Supervised ML	Canonical correlation system which include inputs as spreading factor,coding rate, bandwidth, and energy consumption and for fuzzy interference system it includes energy consumption, throughput, and delay.	Predicted network efficiency and recommendation for parameters where the canonical correlation results which identify relation between independent variables(SF,CR ,and BW)and dependent variables(EC,delay ,throughput and PDR)and for fuzzy system it predicts network efficiency based on these relations.	Obtaining high network effeciency.

Table 3.2: Machine Learning Algorithms(2)

[2]	A supervised ML where an artificial neural network was built and trained to predict received power then particle swarm algorithm was used to find optimal values of parameters corresponding to max received power.The work aims to maximize link performance by optimizing received power.A data set is generated for each SF to be utilized in building ANN to predict loRa network perfor-mance.PSO is utilized to optimize loRa parameters on which received power depends.This cause SNR and spectral efficiency improvement.	Particle Swarm Optimization	Supervised ML	Spreading fac-tor,bandwidth, code rate, transmission power, environmental conditions and distance between ED and GW.	Received power at gateway node and updating parameters that maximize power and some link level performance as RSSI,SNR to evaluate effectiveness of the approach	Improving SNR and spectral efficiency.

Table 3.3: Machine Learning Algorithms(3)

[1]	A supervised learning method which outputs a suggested parameter combination for implementation in short time to achieve rapid deployment.It is an attempt to predict performance indicators with a regression model to provide multiple parameters combination.Then all of parameters are sorted based on random forest prediction result.The parameter decision algorithm will return predicted sequence of parameters.	Random Forest	Supervised ML	Antenna type,packet size ,time on air.Priority list of performance indicators such as success rate, transmission time,RSSI,and error rate.	Suggested parameter combination of parameters based on prediction from random forest model and performance indicators for combination of parameters.The root mean square error is conducted for performance indicators with one obtained from experimental data.	Better throughput compared with ADR.

Table 3.4: Machine Learning Algorithms(4)

[7]	A supervised ML called KNN used for resource allocation where similar nodes are near to each other.After learning process unseen data is fed into KNN algorithm and distance is calculated then decision to which class it should be assigned is made according to pre-dominant class of K nearest points.This algorithm indicates that both energy efficiency and reliability improved.	K-nearest neighbours.	Supervised ML	Training data,test data,value of K which represent number of nearest neighbours to consider when making prediction.	Predicted class label for each test data point and distance between test point and its k-nearest neighbours.	Enhancing both efficiency and reliability.

Table 3.5: Machine Learning Algorithms(5)

[15]	FSVM is an extension of the Support Vector Machine (SVM) algorithm that introduces fuzzy membership functions to handle classification problems with noise or outliers more effectively. It aims to construct a classification hyperplane with the maximum margin between classes while considering the fuzzy membership values of samples, which represent the degree to which a sample belongs to a certain class. FSVM assigns different weights to samples based on their importance and employs machine learning techniques for training and learning.	Fuzzy Support Vector Machine	Supervised ML	Training dataset with labeled samples and regularization parameter	Classification hyperplane parameters,classified decision function,and classification accuracy or performance metrics, indicating the effectiveness of the trained FSVM model compared to other classification models.	the proposed algorithm improves the throughput by 34.12(%) and packets the reception rate by 26(%), significantly improving the network throughput and the packets reception rate.

Table 3.6: Machine Learning Algorithms(6)

[19]	The LP-MAB algorithm is a centralized adaptive configuration scheme designed for LoRaWAN. It combines the benefits of the EXP3 (Exploration-Exploitation with Expert Advice) and SE (Successive Elimination) algorithms to manage the transmission parameters of end devices (EDs) in a dynamic environment. The Network Server (NS) acts as the agent, interacting with EDs to learn optimal transmission configurations that maximize Packet Delivery Ratio (PDR) while minimizing Energy Consumption (EC)	LP-MAB (Learning-based Policy for Multi-Armed Bandits)	Reinf-orcement ML	End Devices (EDs), A vector consisting of Spreading Factor (SF), Transmission Power (TP), Carrier Frequency (CF), and Coding Rate (CR) for each action, and Reception of ACK messages from the EDs	Optimal Transmission Parameters, and Action Selection Probabilities	LP-MAB's energy consump-tion outper-forms other algo-rithms(ADR and No ADR) while maintain-ing a relatively high packet delivery rate.

Table 3.7: Machine Learning Algorithms(7)

Chapter 4

Methodology

4.1 Work Flow

This section describes the implementation of six machine learning algorithms on a dataset taken from the OLoRa Simulator to predict new values for spreading factor. Each algorithm is trained on this dataset and tested on three additional datasets. Accuracy measures were also taken for training and testing, along with evaluation metrics like recall, precision, and f1 score for the supervised machine learning algorithms, while silhouette score, and davies-bouldin for the unsupervised machine learning algorithms.

4.2 Dataset Extraction Process

The dataset extraction process from OLoRaSim for machine learning involves capturing four crucial parameters for each simulated node. Initially, the spreading factor (SF) is assigned to nodes based on their proximity to the server, employing the Adaptive Data Rate (ADR) mechanism. Subsequently, the total energy consumed by each node is calculated using a battery model that considers various factors such as packet length, voltage, and current as shown in equations 4.1 and 4.2. Another essential parameter recorded is the number of packets sent by each node throughout the simulation, offering insights into the node's activity level. Additionally, the transmission power (TP) for each node is randomly assigned. To maintain spatial information, the X and Y coordinates of each node are also included in the dataset, facilitating the re-placement of nodes with their updated SFs following the application of machine learning algorithms. This extracted dataset, comprising SF, energy consumption, packet count, transmission power, and spatial coordinates, is then structured into a CSV format for seamless integration into machine learning workflows, enabling the development and evaluation of SF allocation strategies.

$$R_s = SF \times \frac{CR}{2^{SF}} \times BW \times 1000 \tag{4.1}$$

$$E_s = \frac{PLX}{R_s} \times V \times I \tag{4.2}$$

1	run	minR	repetition	Module	Packets Se	SF	TotalEner	TP	X	Y
2	LoRaSim-N	0	0	LoRa.host	1417	7	0.013681	9	-665.206	713.2921
3	LoRaSim-N	0	0	LoRa.host	1426	7	0.013768	7	80.75292	605.4348
4	LoRaSim-N	0	0	LoRa.host	1469	7	0.014183	4	-428.609	468.0541
5	LoRaSim-N	0	0	LoRa.host	1444	7	0.013942	11	-541.714	682.7784

Figure 4.1: Part of Dataset

4.3 Training and Testing Sets

The dataset is separated into two parts: the first is the training set, which is 80 percent (40000 samples), and the second is for testing, which is the remaining 20 percent (10000 samples) as in Figure 4.2. The features columns are columns of transmission power , number of packets sent , and energy consumed by each node , while the target variable to predict is the spreading factor.

```
# Split into training and test sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

Figure 4.2: Training and Testing Sets

4.3.1 Data normalization

In the machine learning algorithms normalization of data is done before training the model which is done by subtracting the mean of a feature from the one feature value and dividing by the standard deviation where by normalizing the features, we can ensure that each feature contributes proportionally to the model's learning process. The model can now learn patterns across all features more effectively, leading to a more accurate representation of the underlying relationships in the data.

4.4 Training and testing supervised learning techniques

4.4.1 KNN

In Figure 4.3 the KNN is trained on 80 percent of data using SKlearn with number of neighbors that is equal to 4.

```
# training the model on training set
knn = KNeighborsClassifier(n_neighbors=4)
knn.fit(X_train, y_train)
```

Figure 4.3: Training KNN

Determining optimal number of neighbors

A graph is plotted betwwen accuracy and number of neighbors and as shown in Figure 4.4 we can see that having 4 neighbors gives highest accuracy. We didn't take 1 or 2 as such small values for neighbors can result in unstable decision boundaries, hence it is not recommended.

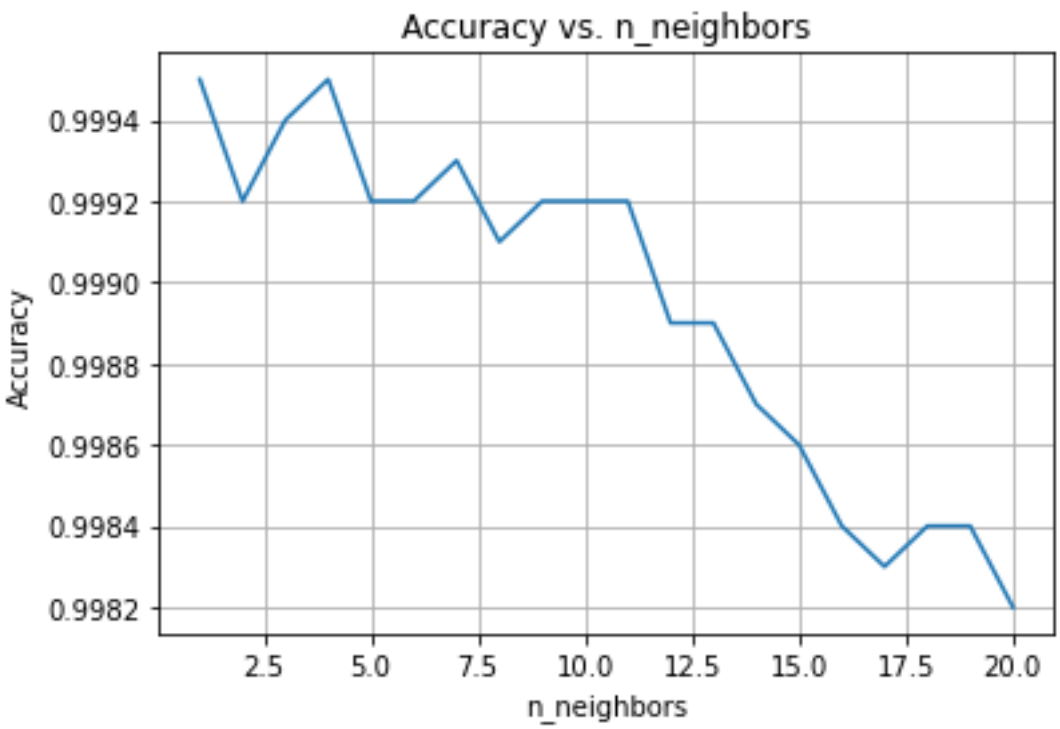

Figure 4.4: KNN Accuracy vs number of neighbors

Testing KNN

The trained model then is tested using SKlearn as in Figure 4.5 on three datasets of sizes 10000, 20000, 30000 and evaluation metrics such as F1 score, precision, recall are calculated for each dataset of them.

```
pred1=knn.predict(X_testt)
print(pred1)
# Calculate the accuracy of the model
accuracypercent1=(knn.score(X_testt, y_testt))*100
print("percentage of accuracy:",accuracypercent1,"%")
precision = precision_score(y_testt, pred1, average='weighted')

# Calculate recall
recall = recall_score(y_testt, pred1, average='weighted')

# Calculate F1-score
f1 = f1_score(y_testt, pred1, average='weighted')
```

Figure 4.5: KNN datasets testing

4.4.2 Naive Bayes

Training Naive Bayes

The same is done for Naive bayes where by using SKlearn the dataset is trained by using 80 percent of it and 20 for testing as shown in Figure 4.6

```
gnb = GaussianNB()
gnb.fit(X_train, y_train)
```

Figure 4.6: Training Naive bayes

Testing Naive Bayes

The trained model is also tested on three datasets as in KNN and also the same evaluation metrics are calculated as in Figure 4.7

```
pred1 = gnb.predict(X_testt)
print("Predicted SF for test data:",pred1)
# Calculate the accuracy of the model
print("Gaussian Naive Bayes model accuracy(in %):",gnb.score(X_testt,y_testt )*100)
precision = precision_score(y_testt, pred1, average='weighted')

# Calculate recall
recall = recall_score(y_testt, pred1, average='weighted')

# Calculate F1-score
f1 = f1_score(y_testt, pred1, average='weighted')
```

Figure 4.7: Testing Naive bayes

4.4.3 Random Forest

Training Random Forest

Here also using Sklearn the random forest is trained using the same 80 percent of the same dataset as in Figure 4.8

```
rf = RandomForestClassifier(n_estimators=100, random_state=42)
rf.fit(X_train, y_train)
```

Figure 4.8: Training Random Forest

Testing Random Forest

Also the same was done on three datasets and same evaluation metrics are determined as in Figure 4.9

```
pred1=rf.predict(X_testt)
print(pred1)
# Calculate the accuracy of the model
accuracypercent=(rf.score(X_testt, y_testt))*100
print("percentage of accuracy:",accuracypercent,"%")
precision = precision_score(y_testt, pred1, average='weighted')

# Calculate recall
recall = recall_score(y_testt, pred1, average='weighted')

# Calculate F1-score
f1 = f1_score(y_testt, pred1, average='weighted')
```

Figure 4.9: Testing Random Forest

4.4.4 Support Vector Machine

Training Support Vector Machine

Using also SKlearn SVM is trained using the same dataset as shown in 4.10

```
svm = SVC(kernel='linear')
svm.fit(X_train, y_train)
```

Figure 4.10: Training SVM

Testing SVM

The same three datasets are used also for testing and same evaluation metrics are calculated as in 4.11

```
pred1 = svm.predict(X_testt)

# Calculate evaluation metrics (same as Random Forest)
accuracypercent = (svm.score(X_testt, y_testt)) * 100
print("Percentage of accuracy for unseen data:", accuracypercent, "%")

precision = precision_score(y_testt, pred1, average='weighted')
recall = recall_score(y_testt, pred1, average='weighted')
f1 = f1_score(y_testt, pred1, average='weighted')
```

Figure 4.11: Testing SVM

4.5 Training and testing unsupervised learning techniques

In this section two unsupervised learning approaches were used K-Means clustering and Gaussian mixture. For each, two evaluation metrics are calculated silhouette score, and davies-bouldin score.

4.5.1 K-Means Clustering

Training K-Means Clustering

The same dataset was used for training (80 percent of it) using SKlearn with number of clusters of 2 as shown in Figure 4.12

```
gmm = GaussianMixture(n_components=2, random_state=42)

# Fit GMM to the training data
gmm.fit(X_train)
```

Figure 4.12: Training K-Means Clustering

Testing K-Means Clustering

As show in Figure 4.13 the accuracy of clustering was evaluated by measuring two evaluation metrics silhouette score and davies bouldin score.

```
silhouette_avg_new = silhouette_score(X_new, new_data_clusters)
davies_bouldin_new = davies_bouldin_score(X_new, new_data_clusters)
```

Figure 4.13: Testing K-Means Clustering

Determining optimal number of clusters

To determine number of clusters to give highest accuracy the elbow plot was used which by iteratively fitting KMeans models with varying numbers of clusters to the dataset, the average distance of data points to their assigned cluster centers is calculated. Plotting these average distance values against the corresponding numbers of clusters reveals a trend wherein the average distance decreases as the number of clusters increases. The Elbow Method entails visually inspecting this plot to identify a point where the rate of decrease in average distance begins to slow down, resembling an "elbow" which as shown in Figure 4.14 it is at 2.

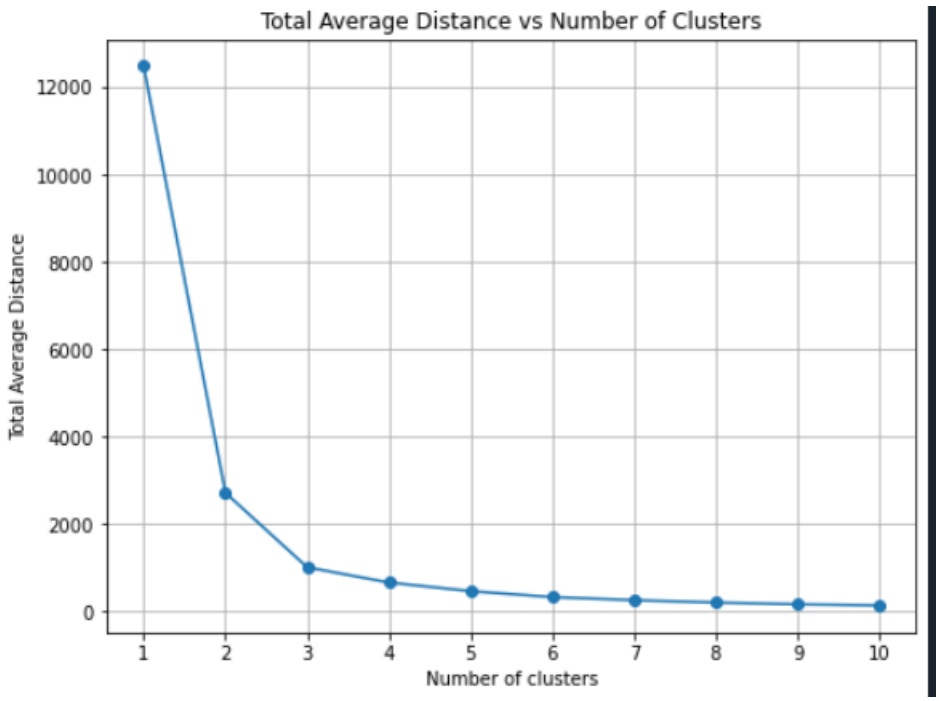

Figure 4.14: Optimal choice of number of clusters

4.5.2 Gaussian Mixture Model

Training GMM

Gaussian mixture model is trained by same dataset with 80 percent of it as in 4.15

```
# Initialize Gaussian Mixture Model (GMM) with 2 compon
gmm = GaussianMixture(n_components=2, random_state=42)

# Fit GMM to the training data
gmm.fit(X_train)

# Predict clusters on the testing data
y_pred = gmm.predict(X_test)
```

Figure 4.15: Training GMM

Testing GMM

The same two evaluation metrics measured in K-Means are measured here also silhouette score and davies bouldin as shown in Figure 4.16

```
silhouette_avg_new = silhouette_score(X_new, new_data_clusters)
davies_bouldin_new = davies_bouldin_score(X_new, new_data_clusters)
```

Figure 4.16: Testing GMM

Chapter 5

Results And Analysis

In this section, the accuracy of each model is assessed using some evaluation metrics. For the supervised learning algorithms KNN, naive bayes, random forest, and support vector machine, three datasets of sizes 10000, 20000, and 30000 are tested, and three evaluation metrics are calculated which are precision, recall, and f1 score.Unsupervised learning techniques, on the other hand, are tested using three datasets of sizes 20000, 30000, and 40000, where silhouette score and davies-bouldin are two evaluation metrics are measured for each of two algorithms, K-Means clustering and GMM.

5.1 Evaluation Metrics

5.1.1 Supervised techniques evaluation metrics

Precision

- Precision measures the proportion of true positive predictions (correctly predicted positive instances) among all instances predicted as positive by the classifier.
- It is calculated as the ratio of true positives to the sum of true positives and false positives.
- Precision ranges from 0 to 1, where a higher precision indicates fewer false positives and better accuracy in positive predictions.

Recall

- Recall measures the proportion of true positive predictions among all actual positive instances in the dataset.
- It is calculated as the ratio of true positives to the sum of true positives and false negatives.

- Recall also ranges from 0 to 1, where a higher recall indicates fewer false negatives and better coverage of positive instances.

F1 Score

- The F1 score is the harmonic mean of precision and recall. It provides a single metric that balances both precision and recall.
- It ranges from 0 to 1, with higher values indicating better overall performance in terms of precision and recall.

5.1.2 Unsupervised techniques evaluation metrics

Silhouette Score

- The silhouette score measures the quality of clusters by assessing how similar an object is to its own cluster compared to other clusters.
- It ranges from -1 to 1, where a high silhouette score indicates that the object is well-matched to its own cluster and poorly matched to neighboring clusters.

Davies Bouldin

- The Davies-Bouldin index evaluates the clustering algorithm's effectiveness by measuring the average similarity between each cluster and its most similar cluster, relative to the cluster's size.
- It ranges from 0 to infinity, where lower values indicate better clustering. A value of 0 indicates perfectly separated clusters.

5.2 Measuring Model Accuracy

5.2.1 Measuring supervised learning techniques accuracy

As shown in tables 5.1, 5.2, 5.3, and 5.4 accuracy and the other three evaluation metrics are calculated for each algorithm(Precision, Recall, and F1 score)including original dataset(10 percent for testing) and three other datasets of sizes 10000, 20000, and 30000

Dataset	Accuracy (%)	Precision	Recall	F1 Score
Dataset 50000	99.88	0.998805	0.9988	0.99879
Dataset 10000	99.4	0.99374	0.9937	0.99372
Dataset 20000	99.86	0.998611	0.9986	0.9986011
Dataset 30000	99.85	0.99374	0.9937	0.99372

Table 5.1: Evaluation Metrics for Different Datasets Using KNN

Evaluation metrics for KNN

Evaluation metrics for Naive Bayes

Dataset	Accuracy (%)	Precision	Recall	F1 Score
Dataset 50000	99.9	0.99879	0.9988	0.9987988
Dataset 10000	99.31	0.993142	0.9931	0.9931
Dataset 20000	99.82	0.99822	0.9982	0.998202
Dataset 30000	99.823	0.998239	0.998233	0.998235

Table 5.2: Evaluation Metrics for Different Datasets Using Naive Bayes

Evaluation metrics for Random Forest

Dataset	Accuracy (%)	Precision	Recall	F1 Score
Dataset 50000	99.99	0.99994	0.99994	0.99993
Dataset 10000	99.34	0.993429	0.9934	0.993413
Dataset 20000	99.82	0.99822	0.9982	0.99821
Dataset 30000	99.83	0.99827	0.99826	0.998268

Table 5.3: Evaluation Metrics for Different Datasets Using Random Forest

Evaluation metrics for Support Vector Machine

Dataset	Accuracy (%)	Precision	Recall	F1 Score
Dataset 50000	99.99	0.99999	0.99998	0.99995
Dataset 10000	99.31	0.99314	0.9931	0.993118
Dataset 20000	99.82	0.9982	0.9982	0.9982
Dataset 30000	99.823	0.9982	0.9982	0.9982

Table 5.4: Evaluation Metrics for Different Datasets Using SVM

5.2.2 Measuring Unsupervised learning techniques accuracy

As shown in tables 5.5,and 5.6 two evalution metrics were measeured for each algorithm(Silhouette score and Davies-bouldin) including original dataset(10 percent for testing) and three other datasets.

Evaluation metrics for K-Means Clustering

Dataset	Silhouette Score	Davies Bouldin
Dataset 50000	0.67891	0.4149
Dataset 20000	0.6634	0.4364
Dataset 30000	0.6788	0.4165
Dataset 40000	0.6815	0.4149

Table 5.5: Evaluation Metrics for Different Datasets K-Means Clustering

Evaluation metrics for Gaussian Mixture Model

Dataset	Silhouette Score	Davies Bouldin
Dataset 50000	0.67456	0.4038
Dataset 20000	0.6718	0.4083
Dataset 30000	0.6730	0.40676
Dataset 40000	0.6745	0.40673

Table 5.6: Evaluation Metrics for Different Datasets Using GMM

5.3 Analysis

5.3.1 Results analysis for supervised techniques

The evaluation metrics for different datasets using KNN, Naive Bayes, Random Forest, and SVM algorithms reveal distinct patterns in performance across varying dataset sizes. For KNN, the accuracy remains consistently high, with a slight dip as the dataset size decreases, showing the highest accuracy of 99.88% for the 50,000 record dataset. The precision, recall, and F1 scores also follow a similar trend, maintaining near-perfect values but slightly lowering with smaller datasets. Naive Bayes demonstrates a consistent performance with high accuracy and F1 scores across all datasets, peaking at 99.9% accuracy for the largest dataset. However, it shows a noticeable drop in accuracy and related metrics with the 10,000 record dataset, indicating some sensitivity to dataset size. Random Forest outperforms the other algorithms with nearly perfect accuracy of 99.99% on the largest dataset, showing robustness across different dataset sizes with minimal

performance variance. SVM also shows exceptional accuracy, matching Random Forest's top performance on the largest dataset but displaying more stability than Naive Bayes across smaller datasets. These results suggest that while all algorithms perform well on large datasets, Random Forest and SVM provide the most consistent and highest performance, even as dataset sizes decrease, indicating their superior robustness and reliability in handling varying amounts of data.

5.3.2 Results analysis for unsupervised techniques

Silhouette and Davies Bouldin Scores

The evaluation metrics for K-Means Clustering and Gaussian Mixture Model contribute to an understanding of how unsupervised learning techniques perform across different datasets. Silhouette score allows for the assessment of the quality of clustering, where a higher score implies good separation of clusters and low intra-cluster distance. In both K-Means Clustering and GMM, the silhouette scores fall in a similar range between about 0.67 and 0.68. That means reasonably good clustering for most data sets with different sizes. Similarly, the Davies Bouldin index indicates the effectiveness of clustering using the measurement of cluster similarity relative to its size, where lower values indicate better clustering. Davies Bouldin index for K-Means Clustering and GMM also fall in a similar range between about 0.40 and 0.44, implying quite low scatter within clusters and quite good separation of clusters. Such results indicate that K-Means Clustering and GMM both fare pretty nicely across different-sized data sets, thus demonstrating their capability to draw meaningful patterns and structures from the data. Still, the slight differences in the performance across the data sets emphasize the role of dataset characteristics in unsupervised learning. Overall, these findings provide worth-having information about how good K-Means Clustering and GMM are at this task of clustering and how they could also be used in multiple applications in data analysis and pattern detect.

5.3.3 Graphical Representation

Number Of Nodes vs Throughput

Figure 5.1 (Matlab graph) illustrates the relationship between the number of users and the throughput achieved by various Spreading Factor (SF) assignment techniques, including Adaptive Data Rate (ADR), KMeans Clustering, KNN, and Naive Bayes. Among these approaches, KMeans Clustering emerges as the most efficient, attaining a throughput of 58.7(%) with 10,000 users. Following closely is Naive Bayes, yielding a throughput of approximately 50%, while ADR exhibits a throughput of around 47%. In contrast, KNN demonstrates the lowest throughput, standing at approximately 44%. With an increase in the number of users from 10,000 to 50,000, the throughput of each technique

declines. Notably, at 20,000 users, KNN and Naive Bayes share identical throughputs of roughly 33%, maintaining parity throughout. At 15,000 users, ADR and Naive Bayes display equivalent throughputs of approximately 45%, after which ADR sustains a superior throughput until reaching 50,000 users.

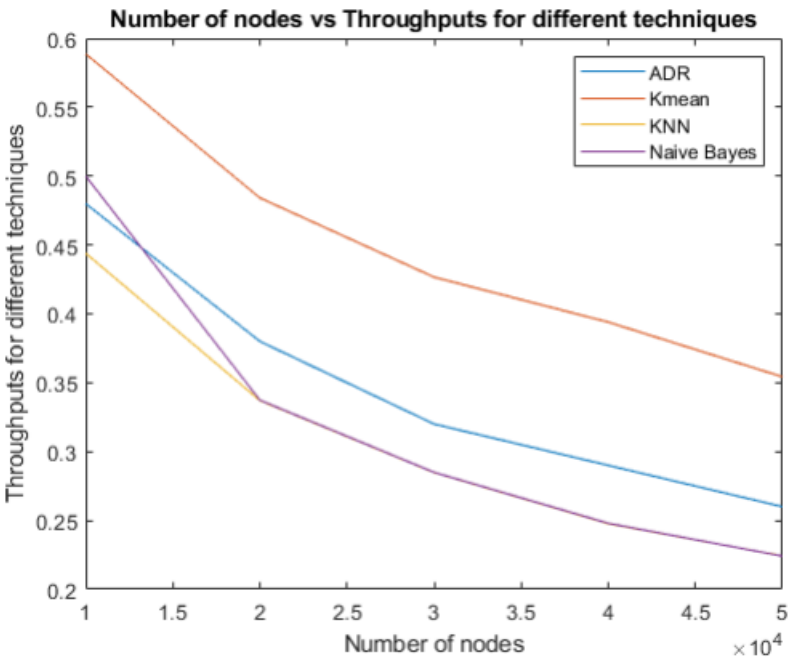

Figure 5.1: Number Of Nodes vs Throughput

Number Of Nodes vs Success Rate

In Figure 5.2 (Matlab graph), the success rates achieved by each SF assignment technique are depicted. Notably, KMeans Clustering exhibits the highest success rate, reaching 9.08. Following closely is Naive Bayes, with a success rate approximately equal to 7.5, while ADR ranks third with a success rate of around 6.5. Conversely, KNN demonstrates the lowest success rate, standing at 6.38. At 20,000 users, KNN and Naive Bayes share identical success rates, while at approximately 14,000 users, Naive Bayes and ADR display equivalent success rates before ADR begins to outperform. As the number of users increases, the success rates for all techniques show an upward trend.

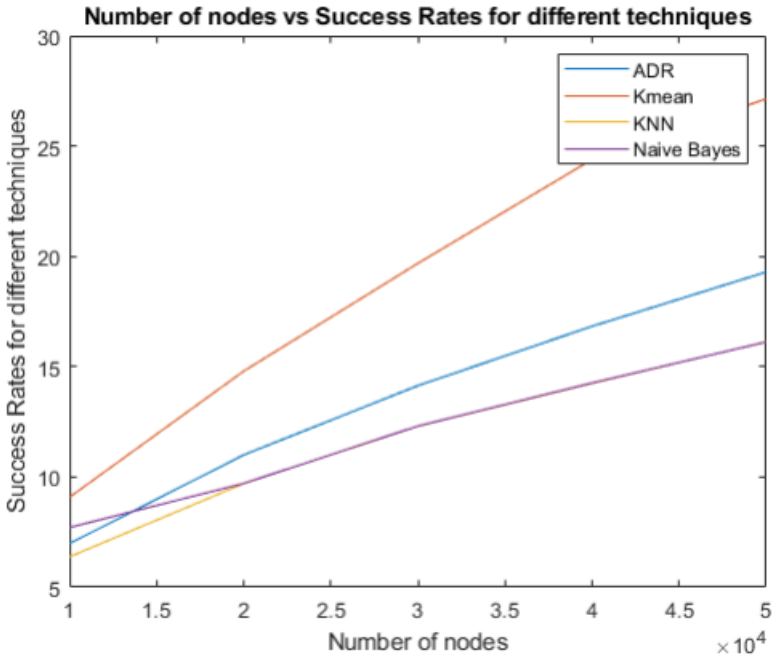

Figure 5.2: Number Of Nodes vs Success Rate

Chapter 6

Conclusion And Future Work

6.1 Conclusion and Future Work

In this study, we looked into the use of several machine learning approaches to optimize LoRa networks, which are critical components of IoT infrastructure. We gained valuable insights into the performance and potential of supervised and unsupervised algorithms, including KNN, Naive Bayes, Random Forest, Support Vector Machine, K-Means Clustering, and the Gaussian Mixture Model. Our results indicate that supervised learning algorithms have strong classification capabilities, indicated by high levels of accuracy, precision, recall, and F1 scores on different datasets. Further, the ability of unsupervised algorithms to segment and cluster data, shown by metrics like silhouette score or Davies-Bouldin index, is success in enhancing network resource allocation as well as overall system performance optimization. The thorough methodology used in our evaluation, which includes training, testing, and analysis, ensures the validity and trustworthiness of our findings, emphasizing the necessity of empirical validation in the development of machine learning solutions for practical applications. These findings add to the expanding body of research on using artificial intelligence to handle the unique issues faced by IoT networks, particularly low-power, wide-area communication technologies such as LoRa. Hybrid approaches that combine supervised and unsupervised methods can be studied as part of the future work in this field. This is to harness the strengths of both systems to enhance and optimize LoRa for a variety of IoT applications. Moreover, when these algorithms are deployed under real-world conditions, their performance will be validated, and the results obtained may provide insights that are helpful as far as their adoption and integration into operational IoT systems are concerned.

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